WV_Thesis_Summary

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Read call data, remove N/A values and separate by date call was listed

```
calls <- read.csv("SPX_ATM_calls_MayJul2024.csv") %>%
  na.omit() %>%
  mutate(
    quote_date = as.Date(date), .before = expiration_date,
    expiration_date = as.Date(expiration_date)
  ) %>%
  select(-date)
```

Obtain daily log returns of SPX from May 2023 to August 2025, add column for 21-trading-day rolling volatility, and annualized using `sqrt(252)`.

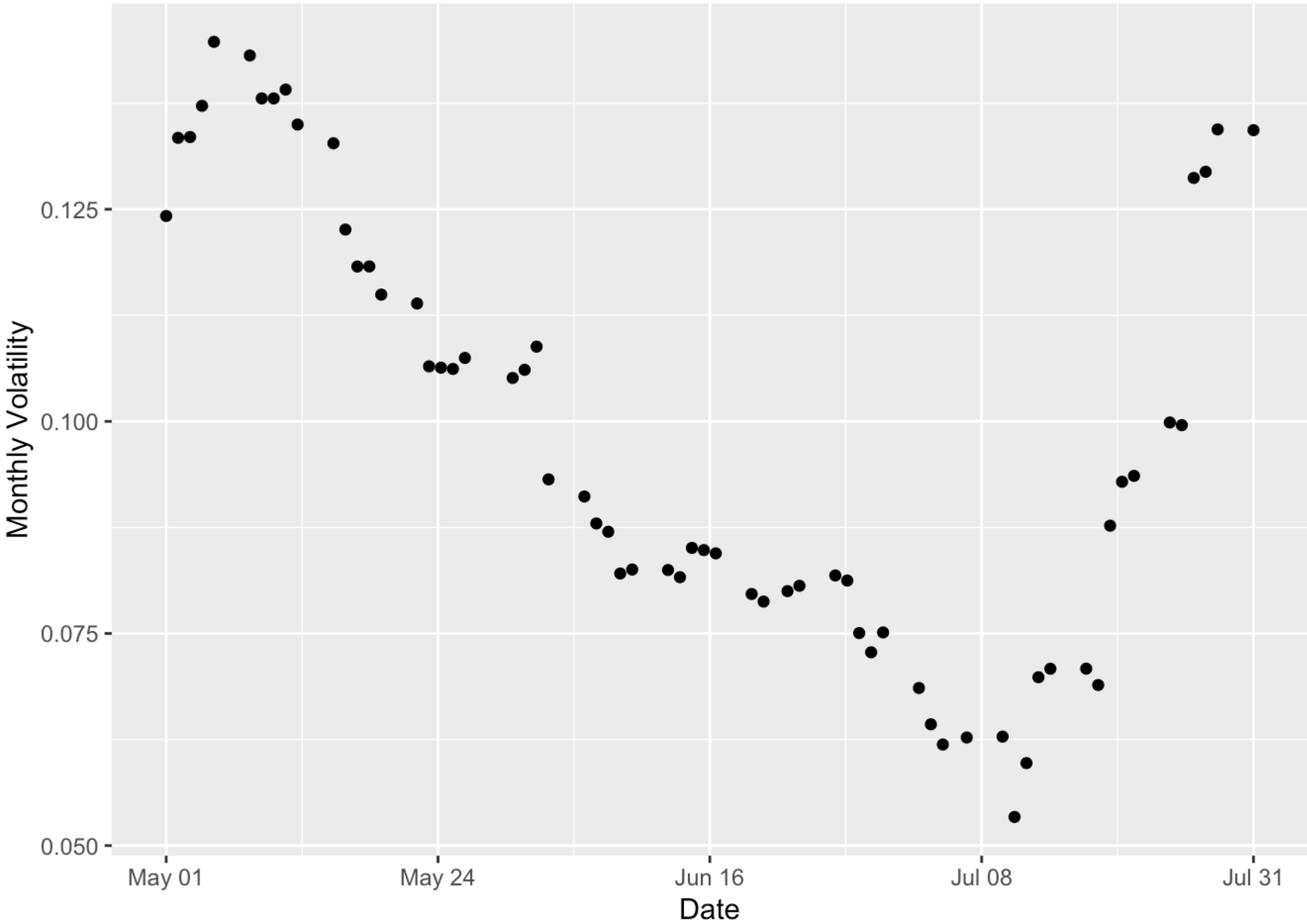
```
spx <- tq_get("^GSPC", get = "stock.prices",
              from = "2024-03-01",
              to = "2024-08-01")

returns <- spx %>%
  tq_transmute(select = adjusted,
              mutate_fun = periodReturn,
              period = "daily",
              col_rename = "daily_return") %>%
  mutate(log_return = log1p(daily_return)) %>%
  mutate(
    roll_sd = slide_dbl(log_return, sd, .before = 21, .complete = TRUE),
    mth_vol = roll_sd * sqrt(252)
  ) %>%
  as_tibble()
```

Filter volatilities or calls within a given range

```
tot_vol <- function(start_date, end_date) {
  start_date <- as.Date(start_date)
  end_date <- as.Date(end_date)

  res <- returns %>%
    filter(date >= start_date & date <= end_date) %>%
    mutate(
      x_day = as.numeric(date - start_date)
    )
  return(res)
}

tot_call <- function(start_date, end_date) {
  start_date <- as.Date(start_date)
  end_date <- as.Date(end_date)

  res <- calls %>%
    filter(expiration_date >= start_date & expiration_date <= end_date) %>%
    mutate(
      x_day = as.numeric(expiration_date - start_date)
    )
  return(res)
}
```

Plot all volatilities

```
tot_vols <- tot_vol("2024-04-27", "2024-07-29")

# Build x-axis breaks
  x_min    <- min(tot_vols$x_day)
  x_max    <- max(tot_vols$x_day)
  x_breaks <- seq(x_min, x_max, length.out = 5)
  x_labels <- format(as.Date(tot_vols$date[1]) + round(x_breaks), "%b %d")

# Build the plot
ggplot(tot_vols, aes(x = x_day, y = mth_vol)) +
  geom_point() +
  scale_x_continuous(
    breaks = x_breaks,
    labels = x_labels
  ) +
  labs(
    x = "Date",
    y = "Monthly Volatility"
  ) +
  theme(plot.title = element_text(hjust = 0.5))
```



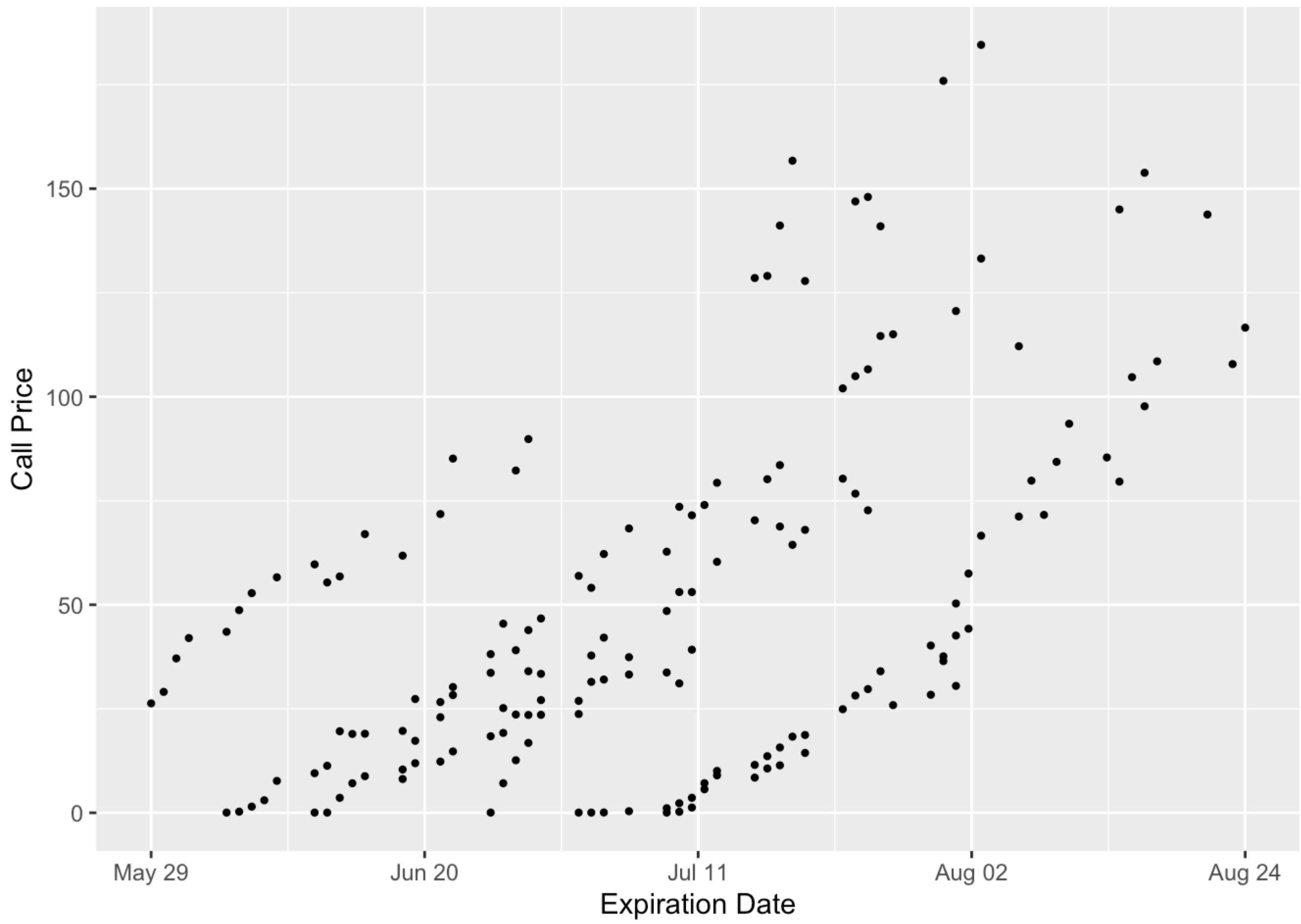
Summary stats for all volatilities

```
tot_vols_tbl <- tibble(
  Obs = nrow(tot_vols),
  Min = min(tot_vols$mth_vol),
  Max = max(tot_vols$mth_vol),
  Mean = mean(tot_vols$mth_vol),
  Median = median(tot_vols$mth_vol),
  "Std Dev" = sd(tot_vols$mth_vol)
) %>% signif(4)

tot_vols_tbl
```

```
## # A tibble: 1 × 6
##     Obs    Min   Max   Mean Median `Std Dev`
##   <dbl>  <dbl> <dbl>  <dbl>  <dbl>     <dbl>
## 1    63 0.0534 0.145 0.0983 0.0929    0.0258
```

Plot all call prices

```
tot_calls <- tot_call("2024-05-27", "2024-08-29")

# Build x-axis breaks
  x_min    <- min(tot_calls$x_day)
  x_max    <- max(tot_calls$x_day)
  x_breaks <- seq(x_min, x_max, length.out = 5)
  x_labels <- format(as.Date(tot_calls$expiration_date[1]) + round(x_breaks), "%b %
d")

# Build the plot
ggplot(tot_calls, aes(x = x_day, y = close)) +
  geom_point(size = 0.8) +
  scale_x_continuous(
    breaks = x_breaks,
    labels = x_labels
  ) +
  labs(
    x = "Expiration Date",
    y = "Call Price"
  ) +
  theme(plot.title = element_text(hjust = 0.5))
```



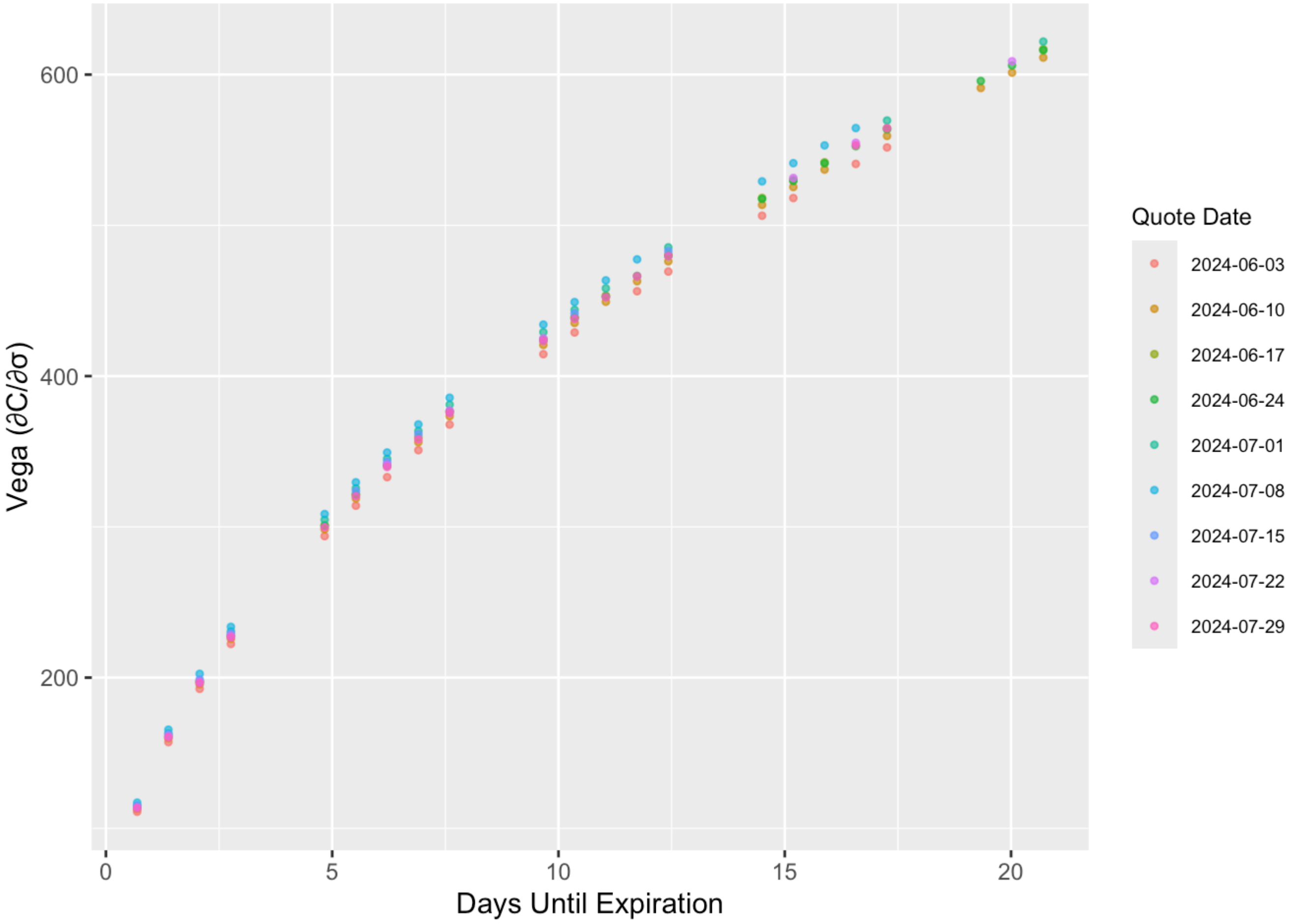
Summary stats for all call prices

```
tot_calls_tbl <- tibble(
  Obs = nrow(tot_calls),
  Min = min(tot_calls$close),
  Max = max(tot_calls$close),
  Mean = mean(tot_calls$close),
  Median = median(tot_calls$close),
  "Std Dev" = sd(tot_calls$close)
) %>% signif(4)

tot_calls_tbl
```

```
## # A tibble: 1 × 6
##     Obs   Min   Max  Mean Median `Std Dev`
##   <dbl> <dbl> <dbl> <dbl>  <dbl>     <dbl>
## 1   167  0.05  185.  50.0   37.6      42.8
```

Compute and plot vega across all call observations

```
# 1. Build a lookup of spot volatility by date from the returns series
vol_lookup <- returns %>%
  select(date, mth_vol) %>%
  filter(!is.na(mth_vol))

# 2. Join spot vol onto each call by its quote_date, then compute vega
vega_dat <- tot_calls %>%
  left_join(vol_lookup, by = c("quote_date" = "date")) %>%
  filter(!is.na(mth_vol)) %>%
  mutate(
    t   = maturity_yrs,
    vol = mth_vol,
    r   = interest_rate / 100,
    d1  = (log(spot_price / strike_price) + (r + 0.5 * vol^2) * t) / (vol * sqrt(t)),
    vega = spot_price * dnorm(d1) * sqrt(t)
  )

# Plot vega vs. time to expiration, coloured by quote date
vega_dat %>%
  filter(is.finite(vega), t > 0) %>%
  ggplot(aes(x = t * 252, y = vega, color = as.factor(quote_date))) +
  geom_point(size = 0.9, alpha = 0.7) +
  labs(
    x     = "Days Until Expiration",
    y     = "Vega (∂C/∂σ)",
    color = "Quote Date"
  ) +
  theme(
    plot.title   = element_text(hjust = 0.5),
    legend.title = element_text(size = 9),
    legend.text  = element_text(size = 7)
  )
```


Vega (∂C/∂σ)
600
400
200
0
5
10
15
20
Days Until Expiration
Quote Date
2024-06-03
2024-06-10
2024-06-17
2024-06-24
2024-07-01
2024-07-08
2024-07-15
2024-07-22
2024-07-29